

TestNG + POM Hybrid Framework

Selenium Test Cases with Data Driven (Excel)

Application	OrangeHRM Demo – https://opensource-demo.orangehrmlive.com
Credentials	Admin / admin123
Test Data File	TestData.xlsx (4 sheets: Login, SearchUser, JobTitle, MultiWindow)
Framework	POM + TestNG + Data Driven (Apache POI) + Hybrid
Total Test Cases	4 (TC01 to TC04)

TC01 – Login with Multiple Credentials (Data Driven)

Concepts Covered:

Explicit Wait • @DataProvider • Excel Read (Apache POI) • Assert • Navigation • TakesScreenshot

1	Open browser and navigate to https://opensource-demo.orangehrmlive.com/web/index.php/auth/login
2	Read test data from Login sheet in TestData.xlsx using Apache POI
3	Load data into @DataProvider — rows: valid credentials, invalid username, invalid password, empty fields
4	Wait explicitly until Username field is visible (ExpectedConditions.visibilityOfElementLocated)
5	Verify page title contains 'OrangeHRM'
6	Clear username field and enter value from Excel column username
7	Clear password field and enter value from Excel column password
8	Click Login button
9	Wait explicitly for either success message or error message to appear
10	Capture actual message displayed on screen using getText()
11	Assert actual message equals expected message from Excel column expectedMessage
12	If login is successful verify URL contains /dashboard
13	If login is successful click Logout and verify redirect to login page
14	Repeat steps 4–13 for all rows in the Excel sheet (4 data sets)
15	Take screenshot after each iteration and save with row number in filename

Excel Test Data – Login Sheet:

username	password	expectedMessage
Admin	admin123	Welcome to OrangeHRM
wronguser	admin123	Invalid credentials

Admin	wrongpass	Invalid credentials
(empty)	(empty)	Required

TC02 – Search Users with Multiple Filter Combinations (Data Driven)

Concepts Covered:

Custom Dropdown • WebTable • findElements • List • @DataProvider • Excel Read • Loop

1	Open browser and login with Admin / admin123
2	Read test data from SearchUser sheet in TestData.xlsx using Apache POI
3	Load into @DataProvider — rows: different Role and Status combinations
4	Navigate to Admin → User Management → Users
5	Wait until System Users page loads completely
6	For each data set from Excel sheet:
7	Locate Role dropdown and select value from Excel column role (custom dropdown – click and select)
8	Locate Status dropdown and select value from Excel column status
9	Click Search button
10	Wait until search results table is fully loaded
11	Get all rows using findElements(By.xpath("//div[@role='row']"))
12	Loop through each row and print Username, Role, Status of each record
13	Verify every row status matches the status value from Excel
14	Get total record count and assert it is greater than 0
15	Assert actual record count matches expectedCount column from Excel
16	Click Reset and verify all filter fields are cleared before next iteration

Excel Test Data – SearchUser Sheet:

role	status	expectedCount
Admin	Enabled	>0
ESS	Enabled	>0
Admin	Disabled	>=0
ESS	Disabled	>=0

TC03 – Add Job Title with Multiple Data Sets (Data Driven)

Concepts Covered:

Form Fill • File Upload (sendKeys) • JavascriptExecutor • TakesScreenshot • @DataProvider • Excel Read

1	Open browser and login with Admin / admin123
2	Read test data from JobTitle sheet in TestData.xlsx using Apache POI
3	Load into @DataProvider — rows: different job titles, descriptions and file paths
4	Navigate to Admin → Job → Job Titles
5	Wait until Job Titles list page loads completely
6	For each data set from Excel sheet:
7	Click Add button
8	Wait until Add Job Title form is visible
9	Enter Job Title from Excel column jobTitle using sendKeys()
10	Enter Job Description from Excel column jobDescription using sendKeys()
11	Scroll down to file upload section using JavascriptExecutor (window.scrollTo)
12	Upload file using sendKeys(filePath) — file path read from Excel column filePath
13	Click Save button
14	Wait until success toast message 'Successfully Saved' appears on screen
15	Take screenshot of success state and save file named with jobTitle value
16	Verify newly added job title appears in the list table
17	Repeat for all rows in Excel sheet

Excel Test Data – JobTitle Sheet:

jobTitle	jobDescription	filePath
QA Automation Engineer	Handles automation testing	C:/files/qa.txt
Manual Tester	Handles manual testing	C:/files/manual.txt
Test Lead	Leads the testing team	C:/files/lead.txt
Performance Tester	Handles load testing	C:/files/perf.txt

TC04 – Multi Window and Actions with Multiple URLs (Data Driven)

Concepts Covered:

Actions Class (Hover) • Keys (ESC) • switchTo() Window • JavascriptExecutor • @DataProvider • Excel Read • Window Handle

1	Open browser and login with Admin / admin123
2	Read test data from MultiWindow sheet in TestData.xlsx using Apache POI
3	Load into @DataProvider — rows: different URLs and expected page titles
4	Wait until Dashboard loads completely
5	Use Actions class to hover over profile icon at top right corner
6	Verify dropdown shows options: About, Support, Change Password, Logout
7	Click About and verify modal appears with OrangeHRM version information
8	Close modal using Keys.ESCAPE keyboard event
9	Use JavascriptExecutor to scroll to bottom of Dashboard page (window.scrollTo(0, document.body.scrollHeight))
10	Use JavascriptExecutor to scroll back to top of page
11	For each data set from Excel sheet:
12	Open a new tab using JavascriptExecutor — window.open()
13	Switch to new tab using driver.switchTo().window(handle)
14	Navigate to url from Excel column url
15	Wait until page loads completely
16	Get current tab title using driver.getTitle()
17	Assert tab title equals value from Excel column expectedTitle
18	Close new tab using driver.close()
19	Switch back to OrangeHRM tab using driver.switchTo().window(originalHandle)
20	Verify OrangeHRM Dashboard is still active after each iteration
21	After all iterations click Logout from profile dropdown and verify redirect to Login page

Excel Test Data – MultiWindow Sheet:

url	expectedTitle
https://google.com	Google
https://github.com	GitHub
https://selenium.dev	Selenium
https://testng.org	TestNG

Selenium Concepts Coverage Matrix

Concept	TC01	TC02	TC03	TC04
Explicit Wait (ExpectedConditions)	✓	✓	✓	✓
Excel Read – Apache POI	✓	✓	✓	✓
@DataProvider (TestNG)	✓	✓	✓	✓
Assert (TestNG)	✓	✓	✓	✓
TakesScreenshot	✓	✗	✓	✗
Custom Dropdown Handling	✗	✓	✗	✗
WebTable + findElements Loop	✗	✓	✗	✗
JavascriptExecutor	✗	✗	✓	✓
File Upload via sendKeys	✗	✗	✓	✗
Actions Class (Hover)	✗	✗	✗	✓
Multi Window Handle	✗	✗	✗	✓
switchTo() Window	✗	✗	✗	✓
Keys – Keyboard Events	✗	✗	✗	✓
Navigation (back/forward)	✓	✗	✗	✗

POM Structure Required: LoginPage.java • DashboardPage.java • UserManagementPage.java • JobTitlePage.java • MyInfoPage.java |
 BaseTest.java • ExcelUtils.java • ConfigReader.java • ScreenshotUtils.java • TestListener.java • testng.xml